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$$\begin{aligned} f(x) &= \operatorname{Bgtan}\left(\frac{x}{a}\right) \text{ met } a \in \mathbb{R}_0 \\ \Rightarrow f'(x) &= \frac{1}{1 + \frac{x^2}{a^2}} \left(\frac{x}{a}\right)' = \frac{1}{1 + \frac{x^2}{a^2}} \frac{1}{a} \\ &= \frac{a^2}{a^2 + x^2} \frac{1}{a} \\ &= \frac{a}{a^2 + x^2} \end{aligned}$$

References